

Harsha S. Bhat

Directeur de recherche CNRS & Professeur attaché
École Normale Supérieure, Paris

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EDUCATION

- 2021 **H.D.R., Supershear Earthquakes** — École Normale Supérieure, France[†]
- 2007 **Ph.D., Mechanical Sciences** — Harvard University, USA *
- 2002 **M.S., Engineering Sciences** — Harvard University, USA
- 2001 **B.E., Civil Engineering** — NITK, India

[†] *Habilitation à Diriger des Recherches* * *Supervised by Prof. James R. Rice & Dr. Renata Dmowska*

CURRENT POSITIONS

- 2024 **Directeur de recherche CNRS** — École Normale Supérieure, France
- 2026 **Professeur attaché** — École Normale Supérieure, France

PAST POSITIONS

- 2022 – 2026 Professeur chargé de cours — École Polytechnique, France
- 2016 – 2024 Chargé de recherche CNRS — École Normale Supérieure, France
- 2021 – 2023 Visiting Professor — NISER, India
- 2012 – 2016 Chargé de recherche CNRS — Institut de Physique du Globe de Paris, France
- 2010 – 2011 Assistant Professor (Research) — University of Southern California, USA
- 2007 – 2010 Post-Doctoral Fellow — University of Southern California, USA
- 2007 – 2010 Visitor in Aeronautics — California Institute of Technology, USA
- 2007 – 2007 Post-Doctoral Fellow — Harvard University, USA
- 2001 – 2007 Graduate Research Associate — Harvard University, USA

FUNDING & GRANTS

* → *waiting for results* gray → *unsuccessful*

- 2026* Agence Nationale de la Recherche – PRCI, France-HongKong coPI
“REFRESH: Revisiting Seismic Hazard and Fault Mechanics in Intraplate Regions under Climatic Stresses”
- 2026* Partenariat Hubert Curien PROCOPE, France-Germany coPI
“CASPER: Climate-Aware Seismic Hazard from Physics-Based Earthquake-Cycle Models”
- 2026 Domaine d’Intérêt Majeur MaTerRE, Île-de-France coPI
“GEO-PROBE: GPU-Accelerated Multi-Physics Spectral Boundary-Element Modeling of Induced Seismicity during Underground Gas Storage”
- 2024 Laboratoire de Recherche Conventionné (CEA), France coPI
“CLIMSEISM: Climate-Triggered Intraplate Seismicity — Detection and Mechanics”
- 2023 Agence Nationale de la Recherche (AAPG 2022), France coPI
“SMEC: Smectite Dehydration — a Key to Fault Mechanics”
- 2023 Partenariat Hubert Curien PROCORE, France-HongKong coPI
“Earthquake Risks along the Anninghe Fault Zone”
- 2022 Partenariat Hubert Curien Aurora, France-Norway coPI
FARM: Fault and Rupture Mechanics
- 2022 CNRS – University of Tokyo, France-Japan coPI
“SESAME: Seasonal Seismicity — Statistical Analysis and Modeling”

2021	Centre National d'Études Spatiales (CNES), France <i>On the Link between Rupture Velocity and Off-Fault Damage Produced in Earthquakes</i>	coPI
2020	Endeavour Fund (MBIE), New Zealand <i>Can Earthquake Simulators Forecast Seismic Hazard after the Next Great Alpine Fault Earthquake?</i>	coPI
2019	European Research Council, Consolidator Grant, EU <i>"PERSISMO: Predicting Energy Release in Fault Systems — Integrating Simulations, Machine Learning and Observations"</i>	PI
2019	Institut National des Sciences de l'Univers (CNRS), TelluS-ALEAS, France <i>IRA: Inverting Rupture Velocity Using Damage and Aftershocks</i>	PI
2019	Marie Skłodowska-Curie ITN (H2020), EU <i>FRICFAULT: Friction in Faults</i>	coPI
2019	Émergence (Ville de Paris), France <i>Inelastic Deformation in the Subduction Zone</i>	coPI
2019	Laboratoire de Recherche Conventionné (CEA), France <i>"Earthquake Dynamics, Damage & Tsunamigenesis"</i>	coPI
2019	Aramont Fund (Harvard), USA <i>"Resolving Earthquake Source Mechanisms on Complex Fault Systems (fellow: K. Okubo)"</i>	coPI
2019	Scheme for Promotion of Academic and Research Collaboration (SPARC), India <i>"Novel Explorations on Earthquake Nucleation within Complex Fault Networks"</i>	coPI
2018	PRESTIGE Grant, France <i>"Seismic Hazard on Thrust Faults Using Physics-Based Models and Surface-Data Analysis (fellow: L. Bruhat)"</i>	host
2018	Thomas Jefferson Fund (FACE Foundation), France–USA <i>Earthquakes in Complex Fault Networks</i>	coPI
2018	École Normale Supérieure (Actions Incitatives), France <i>"Earthquake Mechanics"</i>	PI
2017	Fondation Simone et Cino Del Duca (Académie des sciences), France <i>Dynamics of Fault Networks: Physics-Based Models</i>	PI
2017	Institut National des Sciences de l'Univers (CNRS), mi-lourds, France <i>Observational and Computational Earthquake Cycles</i>	PI
2017	Thomas Jefferson Fund (FACE Foundation), France–USA <i>Earthquakes in Complex Fault Networks</i>	coPI
2016	Agence Nationale de la Recherche, France <i>LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves</i>	PI
2016	Institut National des Sciences de l'Univers (CNRS), TelluS-ALEAS, France <i>"Spontaneous Off-Fault Damage Due to Earthquakes and the Associated Acceleration Spectra"</i>	PI
2016	Institut de Physique du Globe de Paris (Mobilité), France <i>"FDEM Rupture Modeling at Los Alamos (PhD mobility: K. Okubo)"</i>	coPI
2015	Agence Nationale de la Recherche, France <i>LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves</i>	coPI
2015	Émergence (Ville de Paris), France <i>DEED: Deep Earthquake Dynamics</i>	coPI
2015	Natural Environment Research Council (Fellowship), UK <i>Seismic and Aseismic Slip: Properties and Constitutive Behavior of Faults</i>	coPI
2015	Université Sorbonne Paris Cité, France <i>"Earthquake Ruptures on Complex Fault Systems and Complex Media (PhD fellowship K. Okubo)"</i>	PI
2014	European Research Council, Starting Grant (H2020), EU <i>UNITERRA: Unifying On- and Off-Fault Earthquake Models</i>	PI
2014	Émergence (Ville de Paris), France <i>Quantifying the Seismic Hazard Associated with Waste-Water Injection</i>	PI
2014	Natural Environment Research Council, UK <i>Earthquake Fracture Damage and Feedbacks in the Seismogenic Zone</i>	coPI
2014	Agence Nationale de la Recherche, France <i>LIVOGERM: Lithospheric Volcanism and Geothermal Energy at Major Strike-Slip Fault Intersections</i>	coPI

2014	Université Sorbonne Paris Cité, France LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves	coPI
2013	European Research Council, Starting Grant, EU SEISMECH: Towards the Next Generation of Mechanical Models of Earthquake Source Physics	PI
2013	Émergence (Ville de Paris), France GEOSEISMECH: Integrating Mechanical Models with Geological, Experimental and Seismological Observations	PI
2013	AXA Research Fund, France “Seismic Hazard of the Shallow Portion of Megathrusts (postdoc: N. Cubas)”	host
2013	Southern California Earthquake Center (SCEC), USA “Dynamic Formation of Fault-Zone Rocks and Their Influence on Rupture Propagation”	PI
2012	Agence Nationale de la Recherche (Chaires d’Excellence), France GEOSEISMECH: Geological and Seismological Mechanics	PI
2012	Agence Nationale de la Recherche (Blanc, SIMI 6), France GeoSMEC: Geometry of Strike-Slip Faults through Multiple Earthquake Cycles	PI
2011	Émergence (Université Pierre et Marie Curie), France From Damage to Rupture of Quasi-Brittle Materials: Application to Seismic Faults	coPI
2011	National Science Foundation (instrumentation), USA “Optical Interferometer Assembly for Time-Resolved Pointwise Stress & Velocity Measurements in Laboratory Earthquakes”	coPI
2011	Southern California Earthquake Center (SCEC), USA “The Effect of Off-Fault Damage on the Propagation of an Earthquake Rupture”	coPI
2010	National Earthquake Hazards Reduction Program (USGS), USA “Geometric Heterogeneities in Rupture Propagation for the Kunlun–Kunlun Pass–Xidatan Fault System”	PI
2010	National Science Foundation, USA “A Rate-Dependent Micromechanical Damage Mechanics for Simulating Dynamic Earthquake Ruptures”	PI
2010	National Nuclear Security Administration / U.S. Air Force, USA “Generation of High-Frequency P and S Energy by Rock Fracture During a Buried Explosion”	coPI
2008	National Science Foundation, USA Laboratory Earthquakes: Characterization of Ground Motion and Stress States in Complex Rupture Scenarios Using High-Resolution Optical Diagnostics	coPI
2008	National Science Foundation, USA “Micromechanics-Based Modeling of Dynamic Earthquake Rupture in a Structurally Complex Fault Zone”	coPI

HONORS & AWARDS

2018	Grand Prix Michel Gouilloud Schlumberger — French Academy of Sciences
2018	CNRS Award for Doctoral Supervision and Research
2003 – 2006	Harvard University Certificate of Distinction in Teaching (2003, 2004, 2006)

STUDENTS

Postdoctoral Associates

2024 – 2026	Ankit Gupta (India)	
2025 – 2027	Suli Yao (China)	
2022 – 2025	Navid Kheirdast (Iran)	Post-Doctoral Fellow, ISTeP
2021 – 2023	Michelle Almakari (France)	
2021 – 2023	Carlos D. Villafuerte (Mexico)	Asst. Prof., UNAM
2019 – 2022	Ekeabino Momoh (Nigeria)	AXA Postdoc Fellow
2018 – 2021	Lucile Bruhat (France)	Nat. Catastrophe Risk Analyst, AXA
2019 – 2019	Lisa Gordeliy (Russia)	Senior Engineer, ResFrac

2014 – 2016 Marion Y. Thomas (France)

CNRS Scientist, Univ. de Rennes

PhD Students

2024 – 2027 Bharath Shanmugasundaram (India)

2024 – 2027 Yishuo Zhou (China)

2023 – 2026 Thomas Melkior (France)

2023 – 2027 Caiyuan Fan (China)

2021 – 2024 Jinhui Cheng (China)

Postdoc, Caltech

2020 – 2023 Augustin Thomas (France)

Research Scientist, BRGM

2020 – 2023 Joseph Flores Cuba (Peru)

Data Scientist, PowerBI

2018 – 2021 Claudia Hulbert (France)

CEO, Geolabe

2017 – 2020 Samson Marty (France)

Postdoc, Caltech

2014 – 2019 Marshall A. Martinez (USA)

Engineer, Joby Aviation

2015 – 2018 Kurama Okubo (Japan)

Research Scientist, NIED, Japan

2014 – 2017 Pierre Romanet (France)

Research Scientist, Cerema, France

2010 – 2015 Vahe Gabuchian (USA)

Research Scientist, Caltech

2011 – 2014 François X. Passelègue (France)

CNRS Scientist, GeoAzur, Nice

2008 – 2013 Jonathan Mihaly (USA)

Jet Propulsion Laboratory

2007 – 2012 Michael Mello (USA)

Teaching Professor, Caltech

Undergraduate & Masters Interns

2022 Valentin Marnat

2018 Nicolas Mercury

2013 Thibaut Perol

2021 Roxane Ferry

2018 Luc Illien

2012 Lucile Bruhat

2020 Jinhui Cheng

2017 Phillipe Danre

2004 Marion Olives

2019 Roxane Ferry

2015 Eleni Kolokytha

2003 Sonia Fliss

2019 Phillipe Danre

2015 Victor Barolle

2019 Hugo Lestrelin

2014 Kurama Okubo

TEACHING

Current — École Normale Supérieure

Earthquake Source Mechanics | Numerical Methods in Geosciences | Thermodynamics

Past — Harvard, Caltech, IPGP, ENS, École Polytechnique (with various colleagues)

Mécanique des Milieux Continus | Active Faults: Geometry | Seismic Ruptures and Scaling Laws | Introduction to Rock Physics | Mathematical Methods in the Sciences | Environmental Risks and Disasters | Ordinary and Partial Differential Equations | Complex and Fourier Analysis | Computational Solid and Structural Mechanics | Solid Mechanics | Introduction to the Mechanics of Solids | Mechanics of Fracture | Advanced Geomechanics | Mécanique de la Fracturation | Mécanique de la Rupture | Mécanique des Matériaux et des Structures

ORGANIZATION OF SCIENTIFIC MEETINGS

Jan 2026 Earthquake Source: Mechanics, Seismology and Geology Workshop — NISER, India

2023, 2024 Across the Time Scales, from Earthquakes to Earthquake Cycle — EGU

Jun 2019 Coupled Processes in Fracture Propagation in Geo-Materials — CISM Advanced School, Udine

Apr 2015 Multiscale Modeling of Fragmentation & Damage in Fault Zones — SSA

Dec 2014 Fault Zone Properties & Processes During Dynamic Ruptures — AGU

RESPONSIBILITIES & SERVICE

- 2018 – 2024 Team Leader, Faults & Earthquakes Group — ENS
2018 – 2019 Co-organizer, Internal Seminar — ENS
2025 Member, HCERES Committee for ISTerre — Univ. Grenoble
2024 – 2025 Editor, JTCAM

REVIEWING ACTIVITIES

American Geophysical Union | Seismological Society of America | Science | Nature | Nature Communications | Nature Geoscience | Science Advances | Proceedings of the National Academy of Sciences | Journal of the Mechanics and Physics of Solids | European Journal of Mechanics – A/Solids | International Journal of Fracture | Earth and Planetary Science Letters | Geophysical Research Letters | Journal of Structural Geology | Geology | Geophysical Journal International | Journal of Applied Mechanics | Geological Society of America | National Science Foundation | European Research Council

BOOKS

- 2017 Thomas, Marion Y., Thomas M. Mitchell, and H. S. **Bhat**, eds. (2017). *Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture*, Geophysical Monograph 227. American Geophysical Union (AGU). DOI: [10.1002/9781119156895](https://doi.org/10.1002/9781119156895).
2012 Bizzarri, A. and H. S. **Bhat**, eds. (2012). *The mechanics of faulting: From laboratory to earthquakes*. Research Signpost.

BOOK CHAPTERS

- 2022 Thomas, Marion Y. and **Bhat**, Harsha S. (2022a). “Loi de friction et modélisation numérique du cycle sismique”. in *Le Cycle Sismique*. Ed. by Frédérique Rolandone. ISTE-Wiley. DOI: [10.51926/iste.9038.ch4](https://doi.org/10.51926/iste.9038.ch4).
2022 Thomas, Marion Y. and **Bhat**, Harsha S. (2022b). “Friction Laws and Numerical Modeling of the Seismic Cycle”. in *The Seismic Cycle: From Observation to Modeling*. Ed. by Frédérique Rolandone. ISTE-Wiley. DOI: [10.1002/9781394173709.ch4](https://doi.org/10.1002/9781394173709.ch4).

PUBLICATIONS

In Preparation

- 2026 Kheirdast, N, **Bhat**, H. S., Almakari, M, Gupta, A, Villafuerte, C, Thomas, M. Y., and Dubernet, P (2026). “Energy budget of spectrum of slip dynamics emerging from simplified model of fault and damage zone architecture”. [to be subm. J. Geophys. Res.](#)
2026 Garagash, D. I., Brantut, N., **Bhat**, H. S., Schubnel, Alexandre, and Jolivet, Romain (2026). “Low effective stress along faults caused by upwelling fluid flow in laboratory and nature”. [to be subm. J. Geophys. Res.](#)
2026 Gupta, A., **Bhat**, H. S., Faulkner, D. R., Bhattacharya, Pathikrit, and Bollinger, Laurent (2026). “A model for intraplate fault zones using rate and state friction law”. [to be subm. J. Geophys. Res.](#)
2026 Fan, Caiyuan, Lin, Gang, Aubry, Jérôme, Deldicque, Damien, Giorgetti, Carolina, **Bhat**, Harsha S., and Schubnel, Alexandre (2026). “Transition from thermal pressurization to dilatant strengthening during stick-slip ruptures in thermally cracked westerly granite”. [to be subm. J. Geophys. Res.](#)
2026 Momoh, E., Tait, S., and **Bhat**, H. S. (2026). “3-D thermomechanical geodynamic modelling of the French Massif Central, France; and Eifel Volcanic Region, Germany; and prospects for geothermal exploration”. [to be subm. Tectonophysics.](#)

- 2026 Bhagat, R., Sreejith, K., Satriano, C., Gahalaut, V., **Bhat**, H. S., and Bhattacharya, P. (2026). "Aseismic slip and low-frequency earthquakes within an intraplate seismic swarm: the role of fault network geometry". **to be subm.**

Under Review

- 2026 Bhagat, R., Bhattacharya, P., Sreejith, K. M., **Bhat**, H. S., and Gahalaut, Vineet K (2026). "A High-Resolution Earthquake Catalog Reveals Coupled Fault Interactions and Multiple Controls on Seismicity Migration in an Intraplate Earthquake Swarm". **subm. Geophys. Res. Lett.**
- 2026 Zhou, Y., Aochi, H., Schubnel, Alexandre, Ide, S., and **Bhat**, Harsha S. (2026). "Tidal sensitivity of tremors in a mixed fast and slow earthquake system in northeastern Japan". **subm. J. Geophys. Res.** DOI: [10.48550/arXiv.2606.24362](https://doi.org/10.48550/arXiv.2606.24362).
- 2026 Yao, Suli, Yang, Hongfeng, **Bhat**, Harsha S., and Aochi, Hideo (2026). "Supershear Rupture Indicator in Near-fault Particle Motion". **submitted**. DOI: [10.48550/arXiv.2606.10843](https://doi.org/10.48550/arXiv.2606.10843).
- 2026 Thomas, M. Y. and **Bhat**, H. S. (2026). "Une Histoire de la compréhension des Tremblements de Terre". **in Histoire et philosophie des sciences**. Ed. by Yann Vailis. ISTE.
- 2026 Cheng, J., **Bhat**, H. S., Almakari, M., Lecampion, B., and Dubernet, P. (2026). "Quantifying the Role of 3D Fault Geometry Complexities on Slow and Fast Earthquakes". **minor revisions Geophys. Res. Lett.** DOI: [10.48550/arxiv.2602.16403](https://doi.org/10.48550/arxiv.2602.16403).
- 2026 Zhou, Y., Gupta, Ankit, Aochi, H., Schubnel, A., Ide, S., and **Bhat**, H. S. (2026). "Theoretical Constraints on Tidal Triggering of Slow Earthquakes". **minor revisions J. Geophys. Res.** DOI: [10.48550/arxiv.2602.06703](https://doi.org/10.48550/arxiv.2602.06703).
- 2026 Wang, C., Wang, P., Wu, B., **Bhat**, H. S., Bhattacharya, Pathikrit, Xie, Y., Xia, K., and Schubnel, Alexandre (2026). "Complex fault slip behavior modulated by fluid injection rate". **subm. Nat. Comm.**

Published

- 2026 Almakari, Michelle, Kheirdast, N., Villafuerte, C. D., Thomas, M. Y., Dubernet, Pierpaolo, Cheng, Jinhui, Gupta, Ankit, Romanet, Pierre, Chaillat, Stéphanie, and **Bhat**, Harsha S. (2026). "Fault volume digital twin to reproduce the full slip spectrum, scaling and statistical laws". **J. Geophys. Res.** 131, e2025JB032915. DOI: [10.1029/2025JB032915](https://doi.org/10.1029/2025JB032915).
- 2026 Melkior, T., **Bhat**, H. S., and Amlani, Faisal (2026). "Tsunami modeling with dynamic seafloors: a high-order solver validated with shallow water benchmarks". **J. Comput. Phys.** 562.114990. DOI: [10.1016/j.jcp.2026.114990](https://doi.org/10.1016/j.jcp.2026.114990).
- 2026 Michel, Sylvain, Scotti, Oona, Hok, Sebastien, **Bhat**, Harsha S., Kheirdast, Navid, Romanet, Pierre, Almakari, Michelle, and Cheng, Jinhui (2026). "A rate-and-state friction based criterion for the probability of earthquake fault jumps". **J. Geophys. Res.** 131, e2025JB031449. DOI: [10.1029/2025JB031449](https://doi.org/10.1029/2025JB031449).
- 2025 Latour, Soumaya, Lebihain, Mathias, **Bhat**, Harsha S., Twardzik, Cédric, Bletery, Quentin, Hudnut, Kenneth W., and Passelègue, François (2025). "Direct Estimation of Earthquake Source Properties from a Single CCTV Camera". **Science** 390, pp. 463–467. DOI: [10.1126/science.adz1705](https://doi.org/10.1126/science.adz1705).
- 2025 Cheng, Jinhui, Almakari, Michelle, Peruzzo, C., Lecampion, B., and **Bhat**, H. S. (2025). "FASTDASH, a Quasi-dynamic 3D Seismic Cycle Model by Using Boundary Element Method with H-matrices". **Geophys. J. Int.** DOI: [10.1093/gji/ggaf230](https://doi.org/10.1093/gji/ggaf230).
- 2025 Momoh, E., **Bhat**, H. S., Tait, S., and Gerbault, M. (2025). "Volumetric (dilatant) plasticity in geodynamic models and implications on thermal dissipation and strain localization". **Geophys. J. Int.** 240.3, pp. 1551–1578. DOI: [10.1093/gji/ggae463](https://doi.org/10.1093/gji/ggae463).
- 2025 Ferry, Roxane, Thomas, M. Y., **Bhat**, H. S., and Dubernet, Pierpaolo (2025). "Depth Dependence of Co-seismic Off-Fault Damage and its Effects on Rupture Dynamics". **J. Geophys. Res.** e2024JB029787. DOI: [10.1029/2024jb029787](https://doi.org/10.1029/2024jb029787).
- 2024 Petit, Léo, Olive, J-A, Schubnel, Alexandre, Le Pourhiet, L., and **Bhat**, H. S. (2024). "A brittle constitutive law for long-term tectonic modeling based on sub-critical crack growth". **to appear in Geochem. Geophys. Geosyst.** 25, e2023GC011229. DOI: [10.1029/2023gc011229](https://doi.org/10.1029/2023gc011229).
- 2023 Jeandet-Ribes, L., Thomas, M. Y., and **Bhat**, H. S. (2023). "On the importance of setting 3-D stress field in simulations of on- and off-fault deformation". **Geophys. J. Int.** 235.3, pp. 2962–2978. DOI: [10.1093/gji/ggad401](https://doi.org/10.1093/gji/ggad401).

- 2023 Marty, S., Schubnel, A., **Bhat**, H. S., Aubry, J., Fukuyama, E., Latour, S., Nielsen, S., and Madariaga, R. (2023). "Nucleation of laboratory earthquakes: quantitative analysis and scalings". **J. Geophys. Res.** 128.e2022JB026294. DOI: [10.1029/2022jb026294](https://doi.org/10.1029/2022jb026294).
- 2022 Amlani, Faisal, **Bhat**, Harsha S., Simons, Wim J. F., Schubnel, Alexandre, Vigny, Christophe, Rosakis, Ares J., Efendi, Joni, Elbanna, Ahmed, Dubernet, Pierpaolo, and Abidin, Hasanuddin Z. (2022). "Supershear shock front contribution to the tsunami from the 2018 Mw 7.5 Palu, Indonesia earthquake". **Geophys. J. Int.** 230, pp. 2089–2097. DOI: [10.1093/gji/ggac162](https://doi.org/10.1093/gji/ggac162).
- 2021 Jara, Jorge, Bruhat, Lucile, Thomas, M. Y., Antoine, S., Okubo, K., Klinger, Y., Jolivet, Romain, and **Bhat**, H. S. (2021). "Signature of transition to supershear rupture speed in coseismic off-fault damage zone". **Proc. R. Soc. A.** 477, p. 20210364. DOI: [10.1098/rspa.2021.0364](https://doi.org/10.1098/rspa.2021.0364).
- 2021 Elbanna, Ahmed, Abdelmeguid, Mohamed, Ma, Xiao, Amlani, Faisal, **Bhat**, Harsha S., Synolakis, Costas, and Rosakis, Ares J. (2021). "Anatomy of Strike Slip Fault Tsunami Genesis". **Proc. Natl. Acad. Sci. USA.** DOI: [10.1073/pnas.2025632118](https://doi.org/10.1073/pnas.2025632118).
- 2021 **Bhat**, H. S. (2021). "Supershear Earthquakes". *Habilitation à Diriger des Recherches, Ecole Normale Supérieure*.
- 2020 Jeandet-Ribes, Louise, Cubas, Nadaya, **Bhat**, Harsha S., and Steer, Philippe (2020). "Response of a single fault to transient normal stress change, and implications of large erosional events on the seismic cycle". **Geophys. Res. Lett.** 47.e2020GL087631. DOI: [10.1029/2020gl087631](https://doi.org/10.1029/2020gl087631).
- 2020 Jolivet, R, Simons, M, Duputel, Z, Olive, J-A, **Bhat**, H S, and Bletery, Quentin (2020). "Interseismic Loading of Subduction Megathrust Drives Long-Term Uplift in Northern Chile". **Geophys. Res. Lett.** 47.8, e2019GL085377. DOI: [10.1029/2019gl085377](https://doi.org/10.1029/2019gl085377).
- 2020 Okubo, K., Rougier, E, Lei, Z., and **Bhat**, H. S. (2020). "Modeling earthquakes with off-fault damage using the combined finite discrete element method". **J. Comp. Part. Mech.** DOI: [10.1007/s40571-020-00335-4](https://doi.org/10.1007/s40571-020-00335-4).
- 2019 Okubo, K., **Bhat**, H. S., Rougier, E., Marty, S., Schubnel, A., Lei, Z., Knight, E. E., and Klinger, Y. (2019). "Dynamics, radiation and overall energy budget of earthquake rupture with coseismic off-fault damage". **J. Geophys. Res.** 124. DOI: [10.1029/2019jb017304](https://doi.org/10.1029/2019jb017304).
- 2019 Marty, S., Passelègue, François X., Aubry, J., Schubnel, A., **Bhat**, Harsha S., and Madariaga, R. (2019). "Origin of high-frequency radiation during laboratory earthquakes". **Geophys. Res. Lett.** 46. DOI: [10.1029/2018gl080519](https://doi.org/10.1029/2018gl080519).
- 2018 Aubry, J., Passelègue, François X., Deldicque, D., Girault, F., Marty, S., Lahfid, A., **Bhat**, H. S., Escartin, J., and Schubnel, A. (2018). "Frictional heating processes and energy budget during laboratory earthquakes". **Geophys. Res. Lett.** 45. DOI: [10.1029/2018gl079263](https://doi.org/10.1029/2018gl079263).
- 2018 Klinger, Y., Okubo, K., Vallage, A., Champenois, Johann, Delorme, A., Rougier, Esteban, Lei, Zhou, Knight, Earl E., Munjiza, A., Satriano, C., Baize, S., Langridge, Robert, and **Bhat**, Harsha S. (2018). "Earthquake damage patterns resolve complex rupture processes". **Geophys. Res. Lett.** DOI: [10.1029/2018gl078842](https://doi.org/10.1029/2018gl078842).
- 2018 Cruz-Atienza, V. M., Villafuerte, C. D., and **Bhat**, H. S. (2018). "Rapid tremor migration and pore-pressure waves in subduction zones". **Nat. Commun.** 9.1, p. 2900. DOI: [10.1038/s41467-018-05150-3](https://doi.org/10.1038/s41467-018-05150-3).
- 2018 Thomas, Marion Y and **Bhat**, Harsha S (2018). "Dynamic evolution of off-fault medium during an earthquake: a micromechanics based model". **Geophys. J. Int.** 214.2, pp. 1267–1280. DOI: [10.1093/gji/ggy129](https://doi.org/10.1093/gji/ggy129).
- 2018 Romanet, Pierre, **Bhat**, H. S., Jolivet, Romain, and Madariaga, Raül (2018). "Fast and slow earthquakes emerge due to fault geometrical complexity". **Geophys. Res. Lett.** DOI: [10.1029/2018gl077579](https://doi.org/10.1029/2018gl077579).
- 2017 Gabuchian, Vahe, Rosakis, A. J, **Bhat**, H. S., Madariaga, Raül, and Kanamori, Hiroo (2017). "Experimental evidence that thrust earthquake ruptures might open open faults". **Nature** 545.336–339. DOI: [10.1038/nature22045](https://doi.org/10.1038/nature22045).
- 2017 Thomas, M. Y., **Bhat**, H. S., and Klinger, Yann (2017). "Effect of Brittle off-fault Damage on Earthquake Rupture Dynamics". in *Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture*, **Geophysical Monograph 227**. Ed. by M. Y. Thomas, H. S. **Bhat**, and T. M. Mitchell. American Geophysical Union (AGU), pp. 255–280. DOI: [10.1002/9781119156895.ch14](https://doi.org/10.1002/9781119156895.ch14).
- 2017 Passelègue, François X., Latour, Soumaya, Schubnel, Alexandre, Nielsen, Stefan, **Bhat**, Harsha S., and Madariaga, Raül (2017). "Precursory Processes during Laboratory Earthquakes". in *Fault Zone Dynamic Processes : Evolution of Fault Properties During Seismic Rupture*, **Geophysical Monograph 227**. Ed. by

- M. Y. Thomas, H. S. **Bhat**, and T. M. Mitchell. American Geophysical Union (AGU). Chap. 12, pp. 229–242. DOI: [10.1002/9781119156895.ch12](https://doi.org/10.1002/9781119156895.ch12).
- 2016 Perol, T. and **Bhat**, H. S. (2016). “Micromechanics based permeability evolution in brittle materials at high strain rates”. **Pure Appl. Geophys.** Pp. 1–12. DOI: [10.1007/s00024-016-1354-4](https://doi.org/10.1007/s00024-016-1354-4).
- 2016 Passelègue, François X., Schubnel, Alexandre, Nielsen, Stefan, **Bhat**, Harsha S., Deldicque, Damien, and Madariaga, Raül (2016). “Dynamic rupture processes inferred from laboratory microearthquakes”. **J. Geophys. Res.** 121. DOI: [10.1002/2015jb012694](https://doi.org/10.1002/2015jb012694).
- 2016 Mello, M., **Bhat**, H. S., and Rosakis, A. J. (2016). “Spatiotemporal properties of sub-Rayleigh and super-shear rupture velocity fields : Theory and Experiments”. **J. Mech. Phys. Solids** 93, pp. 153–181. DOI: [10.1016/j.jmps.2016.02.031](https://doi.org/10.1016/j.jmps.2016.02.031).
- 2015 Vallage, A, Klinger, Y, Grandin, R, **Bhat**, H. S., and Pierrot-Deseilligny, M (2015). “Inelastic surface deformation during the 2013 Mw 7.7 Balochistan, Pakistan, earthquake”. **Geology** 43.12, pp. 1079–1082. DOI: [10.1130/g37290.1](https://doi.org/10.1130/g37290.1).
- 2015 Frank, W B, Shapiro, N M, Husker, A L, Kostoglodov, V, **Bhat**, H S, and Campillo, M (2015). “Along-fault pore-pressure evolution during a slow-slip event in Guerrero, Mexico”. **Earth Planet. Sc. Lett.** 413, pp. 135–143. DOI: [10.1016/j.epsl.2014.12.051](https://doi.org/10.1016/j.epsl.2014.12.051).
- 2015 Siriki, H., **Bhat**, H. S., Lu, X., and Krishnan, S. (2015). “A Laboratory Earthquake-Based Stochastic Seismic Source Generation Algorithm for Strike-Slip Faults”. **Bull. Seism. Soc. Am.** 105.4, pp. 2250–2273. DOI: [10.1785/0120140110](https://doi.org/10.1785/0120140110).
- 2014 Mello, M., **Bhat**, H. S., Rosakis, A. J., and Kanamori, H. (2014). “Reproducing The Supershear Portion Of The 2002 Denali Earthquake Rupture In Laboratory”. **Earth Planet. Sc. Lett.** 387, pp. 89–96. DOI: [10.1016/j.epsl.2013.11.030](https://doi.org/10.1016/j.epsl.2013.11.030).
- 2013 Passelègue, François X., Schubnel, Alexandre, Nielsen, Stefan, **Bhat**, Harsha S., and Madariaga, Raül (2013). “From Sub-Rayleigh to Supershear Ruptures During Stick-Slip Experiments on Crustal Rocks”. **Science** 340.6137, pp. 1208–1211. DOI: [10.1126/science.1235637](https://doi.org/10.1126/science.1235637).
- 2012 **Bhat**, H. S., Rosakis, A. J., and Sammis, C. G. (2012). “A Micromechanics Based Constitutive Model For Brittle Failure at High Strain Rates”. **J. Appl. Mech.** 79.3. DOI: [10.1115/1.4005897](https://doi.org/10.1115/1.4005897).
- 2011 **Bhat**, H. S., Sammis, C. G., and Rosakis, A. J. (2011). “The Micromechanics of Westerley Granite at Large Compressive Loads”. **Pure Appl. Geophys.** 168.12, pp. 1–18. DOI: [10.1007/s00024-011-0271-9](https://doi.org/10.1007/s00024-011-0271-9).
- 2010 **Bhat**, H. S., Biegel, R. L., Rosakis, A. J., and Sammis, C. G (2010). “The Effect of Asymmetric Damage on Dynamic Shear Rupture Propagation II: With Mismatch in Bulk Elasticity”. **Tectonophysics** 493.3, pp. 263–271. DOI: [10.1016/j.tecto.2010.03.016](https://doi.org/10.1016/j.tecto.2010.03.016).
- 2010 Biegel, R. L., **Bhat**, H. S., Sammis, C. G., and Rosakis, A. J. (2010). “The Effect of Asymmetric Damage on Dynamic Shear Rupture Propagation I: No Mismatch in Bulk Elasticity”. **Tectonophysics** 493.3, pp. 254–262. DOI: [10.1016/j.tecto.2010.03.020](https://doi.org/10.1016/j.tecto.2010.03.020).
- 2010 Mello, M., **Bhat**, H. S., Rosakis, A. J., and Kanamori, H. (2010). “Identifying the unique ground motion signatures of supershear earthquakes: Theory and experiments”. **Tectonophysics** 493, pp. 297–326. DOI: [10.1016/j.tecto.2010.07.003](https://doi.org/10.1016/j.tecto.2010.07.003).
- 2010 Templeton, E. L., **Bhat**, H. S., Dmowska, R., and Rice, J. R. (2010). “Dynamic rupture through a branched fault configuration at Yucca Mountain and resulting ground motions”. **Bull. Seism. Soc. Am.** 100.4, pp. 1485–1497. DOI: [10.1785/012009012110](https://doi.org/10.1785/012009012110). [1785/0120090121](https://doi.org/10.1785/0120090121).
- 2009 Harris, R. A., Barall, M., Archuleta, R. J., Dunham, E. M., Aagaard, B., Ampuero, J-P, **Bhat**, H. S., Cruz-Atienza, V., Dalguer, L., Dawson, P., Day, S. M., Duan, B., Ely, G., Kaneko, Y., Kase, Y., Lapusta, N., Liu, Y., Ma, S., Oglesby, D. D., Olsen, K. B., Pitarka, A., Song, S., and Templeton, E. L. (2009). “The SCEC/USGS dynamic earthquake rupture code verification exercise”. **Seismol. Res. Lett.** 80.1. DOI: [10.1785/gssr1.80.1.119](https://doi.org/10.1785/gssr1.80.1.119).
- 2009 Sammis, C. G., Rosakis, A. J., and **Bhat**, H. S. (2009). “Effects of Off-fault Damage on Earthquake Rupture Propagation: Experimental Studies”. **Pure Appl. Geophys.** 166. DOI: [10.1007/s00024-009-0512-3](https://doi.org/10.1007/s00024-009-0512-3).
- 2009 Templeton, E. L., Baudet, A., **Bhat**, H. S., Dmowska, R., Rice, J. R., Rosakis, A. J., and Rousseau, C. E. (2009). “Finite Element Simulations of Dynamic Shear Rupture Experiments and Dynamic Path Selection Along Kinked and Branched Faults”. **J. Geophys. Res.** B08304. DOI: [10.1029/2008jb006174](https://doi.org/10.1029/2008jb006174).
- 2008 Dunham, E. M. and **Bhat**, H. S. (2008). “Attenuation of radiated ground motion and stresses from three-dimensional supershear ruptures”. **J. Geophys. Res.** 113.B08319. DOI: [10.1029/2007jb005182](https://doi.org/10.1029/2007jb005182).

- 2007 **Bhat**, H. S., Dmowska, R., King, G. C. P., Klinger, Y., and Rice, J. R. (2007). "Off-fault damage patterns due to supershear ruptures with application to the 2001 M_w 8.1 Kokoxili (Kunlun) Tibet earthquake". **J. Geophys. Res.** B06301. DOI: [10.1029/2006jb004425](https://doi.org/10.1029/2006jb004425).
- 2007 **Bhat**, H. S., Olives, M., Dmowska, R., and Rice, J. R. (2007). "Role of fault branches in earthquake rupture dynamics". **J. Geophys. Res.** B11309. DOI: [10.1029/2007jb005027](https://doi.org/10.1029/2007jb005027).
- 2007 **Bhat**, H. S. (2007). "Role of Geometric Complexities and Off-Fault Damage in Dynamic Rupture Propagation". PhD thesis. Harvard University.
- 2005 Fliss, S., **Bhat**, H. S., Dmowska, R., and Rice, J. R. (2005). "Fault branching and rupture directivity". **J. Geophys. Res.** B06312. DOI: [10.1029/2004jb003368](https://doi.org/10.1029/2004jb003368).
- 2004 **Bhat**, H. S., Dmowska, R., Rice, J. R., and Kame, N. (2004). "Dynamic slip transfer from the Denali to Totschunda faults, Alaska: Testing theory for fault branching". **Bull. Seism. Soc. Am.** 94, S202–S213. DOI: [10.1785/0120040601](https://doi.org/10.1785/0120040601).